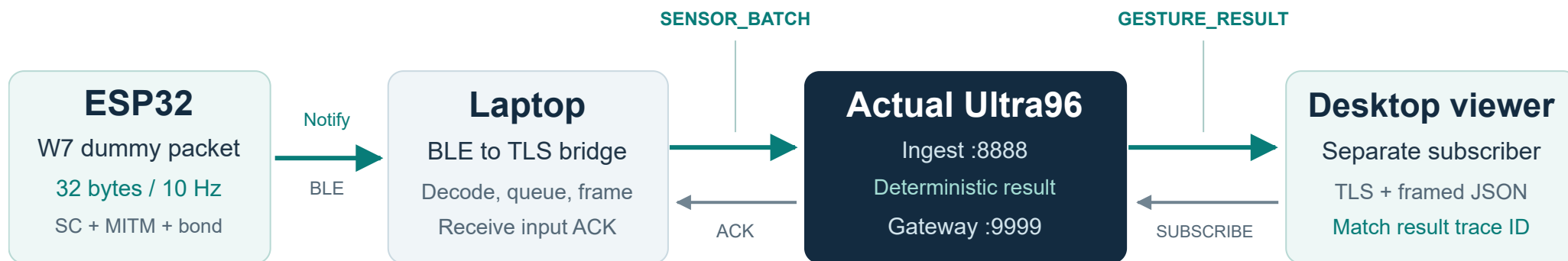


One ESP. A verified connection path.

Real ESP and Ultra96 hardware carry predictable test data to an independent desktop receiver.

Ultra96 external access: TCP 22 (SSH) only

TLS services stay on Ultra96 loopback.



Both application listeners: 127.0.0.1 only

Laptop owns the ingestion tunnel

Laptop 18888 → Ultra96 8888

Both ends bind 127.0.0.1. The bridge opens TLS through this SSH local forward; the Ultra96 returns **INGEST_ACK**.

Viewer owns a separate tunnel

Viewer 19999 → Ultra96 9999

Both ends bind 127.0.0.1. The subscriber opens TLS, sends **SUBSCRIBE**, then receives live **GESTURE_RESULT**.

Each SSH connection reaches Ultra96 port 22 through the institutional jump host. Host keys are verified; TLS validates the demo CA and `ultra96.week7.internal`.

Phone handoff is pending. The final Phone must own its own SSH forward to Gateway 9999. Stop the desktop subscriber first: the latest subscriber replaces the previous one.

Follow one identity across every hop.

W7 v1 has one sample per notification (**N = 1**). At 10 Hz, the payload is exactly **32 bytes**.

0-1 W7 ASCII	2 v1 u8	3 ID u8	4-7 boot_id uint32 LE	8-11 seq uint32 LE	12-15 uptime_ms uint32 LE	16-31 8 signed values 8 × int16 little endian
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Fixed schema: <2sBBIII8h. BLE notification requires ATT MTU ≥ 35; authenticated physical runs measured 517.

ILLUSTRATIVE EXAMPLE / NOT A CAPTURED TRACE

device 1 · boot 7 · seq 42

For channel *i* = 0..7:

```
value[i] = (seq % 2000) - 1000 + 10*i
[-958, -948, -938, -928,
-918, -908, -898, -888]
```

Dummy gesture:

[REST, FIST, OPEN, POINT][seq % 4]
42 % 4 = 2 → **OPEN**, confidence 1.0.

Values are dimensionless sentinels. Confidence is a fixed demonstration field, not measured model accuracy.

Real in this demonstration
ESP hardware, authenticated BLE, independent SSH/TLS, actual Ultra96 and correlated result delivery.

CORRELATION KEY = DEVICE_ID:BOOT_ID:SEQ
1:7:42
Configured session: week7-demo
Boot ID distinguishes device restarts.
OPEN

- 1 Laptop → Ultra96: SENSOR_BATCH**
Decode the packet; preserve device, boot and sequence; attach the configured session.
- 2 Ultra96 → Laptop: INGEST_ACK**
Accepted input returns the same device, boot and sequence. A recent duplicate emits no second result.
- 3 Ultra96 → viewer: GESTURE_RESULT**
Same identifiers, `result_id=1:7:42`, **OPEN**. Compare exact ID sets; ACK and result can arrive in either order.

Dummy by design
Sensor values and gestures. Calibrated sensing, AI/FPGA, two-glove synchronization and AR are outside Week 7.

Show the trace. State the boundary.

Recorded acceptance covers the protected ESP-to-Ultra96 path and the independent desktop subscriber.

ACTUAL REMOTE / 600-SECOND SOAK

5,965 / 5,965

Accepted ACKs / exact matching results.
Zero stream and transport errors.

One BLE/TLS/subscriber connection; no reboot.
One callback discarded at shutdown.

PHYSICAL ACTION FOLLOW-UP

100 + 1,768

Exact matches after reported USB/RESET actions, then a 180-second stability run.

The actions preceded capture. No additional physical interruption was observed during the 180 s run.

BASELINE SOFTWARE CHECKS

181 + 44

Baseline: 181 Python tests passed; 44 portable C# tests passed.

Two Windows skips were Linux-only tests, checked separately. The Unity component was not compiled.

Live show checklist

- 1 Establish provenance.** VPN connected; verify host trust, actual Ultra96 service and loopback listeners.
- 2 Open two independent SSH forwards.** Laptop 18888 → 8888; desktop viewer 19999 → 9999. Both use Ultra96 SSH port 22.
- 3 Subscribe before sending.** Confirm SUBSCRIBED; start the protected BLE bridge and a bounded remote run.
- 4 Point to one shared trace.** Show device, boot, sequence, accepted ACK and matching result ID in the separate streams.
- 5 Finish with counted evidence.** Show exact ID-set match, duration and error/drop metrics; close owned clients and forwards.

Use this pack's README and runbooks for commands. If the board path is unavailable, label an offline walkthrough explicitly.

Pending: actual Phone + Unity

Phone: run on the physical device using its own SSH forward, verify TLS and display, then collect 100-message / 600-second correlation and lifecycle recovery evidence.

Unity: build the supplied component and complete teammate integration on the actual target. Portable C# tests do not establish Unity or Phone acceptance.

Physical faults: separately capture true USB removal/reconnection and the physical RESET-button interruption during streaming.

Recovery evidence already recorded: ingestion-tunnel loss, viewer-tunnel loss, Ultra96 service restart and injected ESP RTS reset. These intentionally interrupted runs remain fault tests, with their losses and unmatched observations preserved.